

# Supervised Machine Learning on Titanic Survival and California Housing Datasets: A Comparative Study

Noah Wiley

*CSCI 164 – Artificial Intelligence Programming – Spring 2026*

California State University, Fresno

noahwiley@mail.fresnostate.edu

Saul Menndoza

*CSCI 164 – Artificial Intelligence Programming – Spring 2026*

California State University, Fresno

nickmendoza24@mail.fresnostate.edu

**Abstract**—This paper presents a comparative study of supervised machine learning algorithms applied to two benchmark datasets: the Titanic survival dataset (binary classification) and the California Housing dataset (regression). Three algorithms are evaluated on each task – Logistic Regression, Decision Tree, and k-Nearest Neighbors – using scikit-learn. Models are trained with standardized pipelines, evaluated using 5-fold cross-validation, and optimized via GridSearchCV. For classification, Logistic Regression achieved the highest accuracy of 81.5% and an AUC of 0.865, consistent with results reported in prior literature. For regression, the tuned Decision Tree Regressor produced the strongest performance. Results are compared against published benchmarks from VanderPlas (2016) and Géron (2019), with differences attributed to variations in preprocessing, random state, and the absence of ensemble methods.

**Index Terms**—supervised learning, classification, regression, scikit-learn, Titanic, California Housing, hyperparameter tuning, GridSearchCV

## I. INTRODUCTION

Machine learning has become a foundational tool for extracting predictive insights from structured data. Classical supervised learning algorithms such as Logistic Regression, Decision Trees, and k-Nearest Neighbors (k-NN) remain widely used due to their interpretability, efficiency, and strong baseline performance across a variety of domains.

This project applies these algorithms to two well-known benchmark datasets: the Titanic survival dataset for binary classification and the California Housing dataset for regression. The objectives are to:

- Explore, preprocess, and analyze each dataset
- Train and evaluate multiple supervised learning models
- Optimize models using systematic hyperparameter search
- Compare results against prior published work
- Reflect on the strengths and limitations of each approach

All experiments are implemented in Python using scikit-learn and run in Google Colab.

## II. DATASETS

### A. Dataset 1: Titanic Survival (Classification)

The Titanic dataset contains records for 891 passengers aboard the RMS Titanic, which sank on April 15, 1912. The binary target variable `survived` indicates whether a passenger survived (1) or did not (0). The overall survival rate in the dataset is 38.4%

Features used include passenger class (`pclass`), sex, age, number of siblings/spouses (`sibsp`), number of parents/children (`parch`), fare, and port of embarkation. Two engineered features were added: `family_size` (sum of `sibsp` and `parch`) and `is_alone` (binary flag for solo travelers).

This dataset was selected due to its mix of numerical and categorical features, the presence of missing values requiring preprocessing, and its widespread use in the machine learning literature, which facilitates meaningful comparison against prior work.

### B. Dataset 2: California Housing Prices (Regression)

The California Housing dataset is derived from the 1990 U.S. Census and contains 20,640 records at the census block level. The continuous target variable `MedHouseVal` represents the median house value in units of \$100,000. Eight numerical features are provided: median income, house latitude, and longitude.

The dataset was selected as a clean regression benchmark with no missing values, sufficient sample size for reliable cross-validation, and well-documented prior results for direct comparison.

## III. DATA PREPROCESSING

### A. Titanic

Missing values in the `age` and `fare` columns were imputed using the column median. Rows with missing embarked values were dropped, resulting in 889 usable records. The `sex` column was binary encoded (male = 0, female = 1)

and embarked was integer encoded ( $S = 0$ ,  $C = 1$ ,  $Q = 2$ ). A stratified 80/20 train/test split was applied to preserve class balance. All models were wrapped in a `Pipeline` with `StandardScaler` to normalize features prior to training.

### B. California Housing

No missing values were present in this dataset. Features were separated from the target variable and an 80/20 train/test split was applied. As with the Titanic dataset, all models used a `Pipeline` with `StandardScaler` to ensure consistent feature scaling across algorithms.

## IV. MODEL DEVELOPMENT

Three supervised learning algorithms were selected for each task to provide a meaningful comparison across linear, tree-based, and distance-based paradigms.

### A. Logistic Regression / Linear Regression

Logistic Regression serves as the linear baseline for the classification task. It models the log-odds of the positive class as a linear combination of input features and is well-suited to problems where the decision boundary is approximately linear. For the regression task, Linear Regression minimizes the sum of squared residuals and provides a fast interpretable baseline.

### B. Decision Tree

Decision Trees recursively partition the feature space based on threshold splits that maximize information gain (classification) or minimize mean squared error (regression). They are capable of capturing non-linear relationships but are prone to overfitting without depth constraints, making them a natural candidate for hyperparameter tuning.

### C. k-Nearest Neighbors

k-NN is a non-parametric algorithm that makes predictions based on the majority class (classification) or average value (regression) of the  $k$  nearest training points by distance. It requires no training phase but is sensitive to feature scale – hence the use of `StandardScaler` – and can be computationally expensive at prediction time on large datasets.

## V. HYPERPARAMETER TUNING

`GridSearchCV` with 5-fold cross-validation was used to optimize the Decision Tree and k-NN models on both datasets. Logistic Regression and Linear Regression were excluded from tuning as they performed competitively with default settings and have fewer meaningful hyperparameters to search.

### A. Decision Tree Parameter Grid

The following parameters were searched:

- `max_depth`: [3, 5, 7, 10, None]
- `min_samples_split`: [2, 5, 10]
- `min_samples_leaf`: [1, 2, 4]

Constraining `max_depth` was found to be the most impactful parameter for reducing overfitting in both the classifier and regressor. The best classification parameters were `max_depth=5`, `min_samples_leaf=1`, and

`min_samples_split=10`, achieving a best CV accuracy of 0.8158. The best regression parameters were `max_depth=10`, `min_samples_leaf=4`, and `min_samples_split=2`, achieving a best CV  $R^2$  of 0.7099.

### B. k-NN Parameter Grid

The following parameters were searched:

- `n_neighbors`: [3, 5, 7, 9, 11, 15]
- `weights`: [uniform, distance]
- `metric`: [euclidean, manhattan]

The best classification parameters were `n_neighbors=9`, `weights=uniform`, and `metric=manhattan`, achieving a best CV accuracy of 0.8115. The best regression parameters were `n_neighbors=9`, `weights=distance`, and `metric=manhattan`, achieving a best CV  $R^2$  of 0.7313.

## VI. RESULTS EVALUATION

### A. Classification Results – Titanic

Table I summarizes the classification metrics for all models. Logistic Regression achieved the highest accuracy (81.5%) and AUC (0.865), indicating strong discriminative ability. The baseline Decision Tree underperformed relative to the other models, consistent with its tendency to overfit without depth constraints. After tuning, the Decision Tree improved noticeably. k-NN demonstrated competitive performance and strong cross-validation consistency, with the tuned variant achieving an AUC of 0.846.

TABLE I  
CLASSIFICATION RESULTS – TITANIC SURVIVAL

| Model                 | Acc   | F1    | AUC   | CV                |
|-----------------------|-------|-------|-------|-------------------|
| Logistic Regression   | 0.815 | 0.744 | 0.865 | $0.792 \pm 0.016$ |
| Decision Tree         | 0.747 | 0.672 | 0.728 | $0.774 \pm 0.021$ |
| k-NN                  | 0.781 | 0.702 | 0.832 | $0.808 \pm 0.020$ |
| Decision Tree (Tuned) | 0.770 | 0.655 | 0.825 | $0.818 \pm 0.029$ |
| k-NN (Tuned)          | 0.787 | 0.721 | 0.846 | $0.799 \pm 0.031$ |

### B. Regression Results – California Housing

Table II summarizes the regression metrics. Linear Regression provided a solid baseline but was outperformed by both the Decision Tree and k-NN, suggesting the presence of non-linear relationships in the data. The baseline Decision Tree showed poor cross-validation performance ( $CV R^2 = 0.355$ ), confirming severe overfitting that was substantially corrected through tuning. The tuned k-NN achieved the strongest overall performance with  $R^2$  of 0.722 and RMSE of 0.603.

TABLE II  
REGRESSION RESULTS – CALIFORNIA HOUSING

| Model                 | MAE   | RMSE  | $R^2$ | CV                |
|-----------------------|-------|-------|-------|-------------------|
| Linear Regression     | 0.533 | 0.746 | 0.576 | $0.553 \pm 0.062$ |
| Decision Tree         | 0.454 | 0.703 | 0.623 | $0.355 \pm 0.085$ |
| k-NN                  | 0.446 | 0.658 | 0.670 | $0.521 \pm 0.044$ |
| Decision Tree (Tuned) | 0.431 | 0.639 | 0.689 | $0.520 \pm 0.103$ |
| k-NN (Tuned)          | 0.406 | 0.603 | 0.722 | $0.562 \pm 0.047$ |

## VII. COMPARISON TO PRIOR WORK

### A. Titanic – VanderPlas (2016)

VanderPlas [2] applies Logistic Regression and Random Forest to the Titanic dataset using comparable preprocessing, reporting Logistic Regression accuracy of approximately 81%. Our result of 81.5% is directly consistent with this benchmark, validating our preprocessing pipeline and model implementation. More competitive Kaggle submissions employing ensemble methods such as Random Forest and XGBoost report accuracies in the 83–86% range [4], representing the performance ceiling achievable beyond classical single-estimator models.

The marginal difference between our result and published ensemble results is expected. Ensembles reduce variance by aggregating predictions across multiple models, an advantage not available to the single-estimator algorithms evaluated here.

### B. California Housing – Géron (2019)

Géron [3] uses the California Housing dataset as a primary regression example, reporting Linear Regression RMSE of approximately 0.72 and a tuned Decision Tree RMSE near 0.71. Our Linear Regression RMSE of 0.746 and tuned Decision Tree RMSE of 0.639 are consistent with these benchmarks. Géron further reports that a Random Forest achieves RMSE near 0.50, illustrating the performance gap between single estimators and ensemble methods. Our tuned k-NN RMSE of 0.603 represents a meaningful step toward that ceiling while remaining within the single-estimator category.

Any remaining differences are attributable to differences in the train/test split seed, the specific hyperparameter grid searched, and minor variations in preprocessing. Both experiments use the same source dataset and scikit-learn implementations, so the results are directly comparable in methodology.

## VIII. DISCUSSION

### A. Strengths

All models were implemented within `Pipeline` objects ensuring that preprocessing steps such as scaling were applied consistently within each cross-validation fold, preventing data leakage. The use of both held-out test sets and 5-fold cross-validation provides a robust estimate of generalization performance. Hyperparameter tuning via `GridSearchCV` systematically improved model performance beyond baseline defaults, most notably correcting the Decision Tree Regressor’s severe overfitting (CV  $r^2$  improving from 0.355 to 0.520).

### B. Limitations

The Titanic dataset is small (891 records), which limits statistical reliability and makes results sensitive to the random seed used for splitting. The California Housing dataset includes latitude and longitude features that encode spatial information which simple distance-based and linear models cannot fully exploit without additional geographic feature engineering. All models evaluated are single estimators; ensemble methods would be expected to outperform all results presented here on both tasks.

### C. Real-World Relevance

Despite their relative simplicity, the models evaluated here have genuine practical relevance. Logistic Regression remains a widely deployed classifier in credit scoring, medical diagnosis, and fraud detection due to its interpretability and calibrated probabilities. Decision Trees underpin many real-world rule-based systems and serve as the building blocks of ensemble methods. k-NN is used in recommendation systems and anomaly detection. The California Housing regression task directly mirrors automated valuation models used in real estate platforms.

## IX. CONCLUSION

This project demonstrated the complete supervised machine learning workflow on two benchmark datasets using scikit-learn. Logistic Regression proved to be the strongest classifier on the Titanic task with 81.5% accuracy and AUC of 0.865, achieving results consistent with published benchmarks. For California Housing regression, non-linear models outperformed the linear baseline, with tuned k-NN achieving the best  $R^2$ . Future work would benefit from exploring ensemble methods, deeper feature engineering, and more exhaustive hyperparameter search strategies to close the gap with state-of-the-art results.

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